Day 11 Report: Reinforcement Learning Trading Bot

PPO agent (stable-baselines3) in a custom Gymnasium trading environment, with backtest

1. Goal

Train an **RL agent** that learns when to **Buy, Sell or Hold** a stock, then test it on a period it never saw and compare it with simple strategies.

2. How it works

RL is learning by trial and error, like training a pet with rewards. The agent looks at the market (state), picks an action, and gets a reward: positive if the money grows, negative if it falls.

RL loop (repeats every day)



3. Data and setup

Item	Value
Data	**Synthetic** price series (Yahoo Finance download failed in Colab), 2,610 days
Train period	1,813 days (2015 to 2021)
Test period	783 days (2022 to 2024), never seen in training
Algorithm	PPO, 150,000 steps, 4.3 min training
Transaction cost	0.1% per trade
Baselines	Buy and Hold (buy once, never sell) and Random actions

4. Results (test period, after costs)

Strategy	Total return %	Sharpe ratio	Max drawdown %	Trades
RL agent (PPO)	-1.6	0.07	-41.3	154
Buy and Hold	+16.8	0.33	-42.8	1
Random	+1.8	0.12	-51.3	275

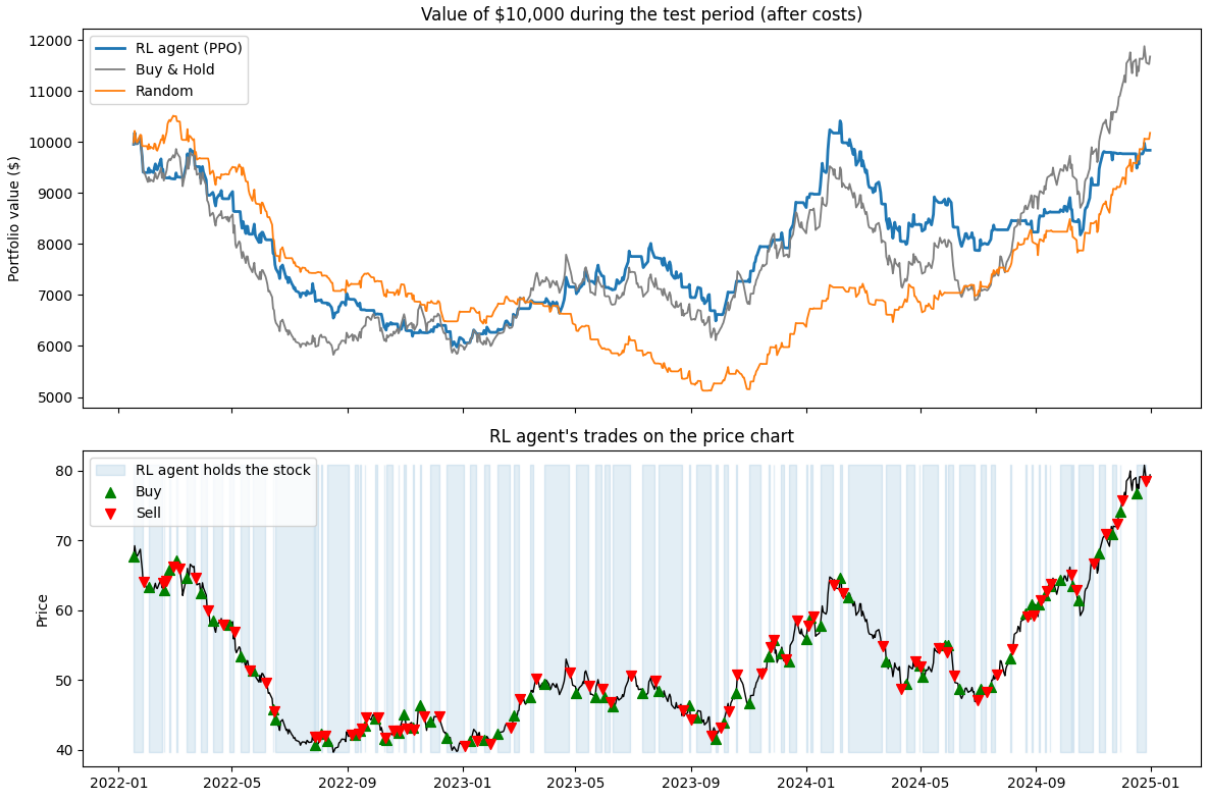


Figure 1. Value of $10,000 over the test period, and the agent's trades

5. Key findings

- **Buy and Hold won** (+16.8%). The RL agent ended slightly negative (-1.6%).
- The agent was **better in falling periods**: in early 2022 and mid-2023 it lost less than Buy and Hold because it was out of the market part of the time.
- It **missed the big rise** at the end of 2024 because it kept selling, so it finished behind.
- 154 trades cost money (0.1% each). Trading too often is a common RL problem.
- It still beat Random on risk (smaller drawdown), so it learned something, but not enough to beat the simple benchmark.

6. Problems and fixes

- yfinance failed in Colab ("Impersonating chrome152 is not supported"). A Stooq CSV fallback was added; this run used the synthetic backup, so results are for practice only.
- PPO warned about running on GPU: fixed by using device="cpu" (faster for small networks).

7. Limitations and next steps

- Re-run with real AAPL data from the Stooq fallback.
- Add a penalty for frequent trading, or train longer, to reduce the 154 trades.
- Try DQN and compare. Note: past backtests do not guarantee future profit.